

Discovering the Metaverses: Towards Open Federated Immersive Authoring Platforms

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ABSTRACT

Social features are critical to contemporary virtual worlds or metaverses. We compare current platforms and motivate a new open-source platform to introduce high school students to authoring immersive media content. Many constraints around privacy, hardware, and licenses constrain the options for using these existing platforms and tools in informal education contexts. We propose that a federated model of servers could support innovation and communities.

Keywords: Metaverse, Open Source, Authoring Platforms, Virtual Reality, Immersive Authoring, Horizon Worlds, Rec Room, VRChat, Mozilla Hubs, Roblox, Zoe Immersive

Index Terms: Human-centered computing—Collaborative and social computing—Collaborative and social computing systems and tools—Open source software

1 INTRODUCTION

Creating an effective set of tools to accomplish immersive authoring for XR content has been often associated with the science fiction trope of the Star Trek holodeck. There has been increased interest from commercial platforms and researchers interested in virtual environments and 3D interaction. It promises to increase participation in creating experiences and content for a larger audience of users. We will focus specifically on virtual reality (VR) platforms in this paper and use “*in situ*” to refer to authoring activities that take place in the same application and interface as the final experience.

Authoring is comprised of two main activities: creating behaviors that dictate how objects and interfaces respond to user input, including any simulations or other calculations, and creating and positioning 3D models and applying textures. Immersive authoring captures some or all these activities and is further described by how the content is constrained for use, such as whether it is available outside the platform it was created on, and how much freedom those tools allow in creating new types of experiences. Many of the experiences we will review have an overall model of a shared virtual space where multiple users can interact with each other by moving their avatars and interacting with objects that have pre-defined behaviors and simulated physics. Some things like points and triggers support a variety of mechanics though not the full spectrum of mechanics and experiences that are possible with a full development environment.

The fact that many of these immersive authoring environments implement social features for not only sharing content created but also creating new content makes it a prime candidate for education.

Learning programming concepts is more effective when done in a pair programming context, especially in XR environments [1]. The platforms discussed in this paper target two types of users: educational users, including institutions and individual teachers who wish to use it in classrooms, and end users for headsets who may own the hardware and wish to use it recreationally and socially.

While we acknowledge the term “metaverse” is used widely to describe a socially engaged community of creators and participants in a persistent VR environment whose promises are further elaborated in [2], we want to also acknowledge that the definition is fluid and we believe there will be innovation and discovery over the next few years that will explore what systems, rules, and approaches will be popular and useful going forward.

This paper addresses the larger contexts of these platforms, how their features support immersive social authoring, and how VR authoring systems such one in development by the authors, RealityFlow may advance the field. We have developed an immersive authoring tool composed of modeling and programming features using the Ubiq networking library to synchronize clients [3]. We conceived of the system while working with secondary education students outside of formal learning contexts, but still within their unique constraints and costs. This paper establishes both how common architectures that use a central server is for XR content platforms and how there is an opportunity for alternative approaches to content sharing. Federation is one option, a model adopted by social media platforms including Mastodon [4], “Your self-hosted, globally interconnected microblogging community.” Users could participate in multiple communities, while developer-users could contribute to more of the platform than just creating content. We describe our efforts to expand an existing immersive authoring tool alongside a case study of a program that solidified our need for open and freely available platform software, concluding with future directions for researchers.

2 USER-GENERATED CONTENT TOOLS AND PLATFORMS

Several tools allow users to create models, scenes, and even experiences inside of VR. These are generally created with end users or formal education as the assumed context. The use case we are addressing includes introducing students in informal learning to VR and creation tools using shared headsets, which precludes most licenses and models where user accounts would be tied to devices or specific teacher/classroom licenses. We envision students having access to a pool of VR devices for checkout and using them to develop and share content amongst other users without worrying about content or users from outside the device-user community.

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Table 1: Immersive authoring platforms

<i>Platform</i>	<i>Web Interfaces</i>	<i>Monetization</i>	<i>Clients</i>	<i>Engine</i>	<i>Scripting</i>	<i>Modeling</i>
Horizon Worlds [8]	Browse, Client	Tokens	VR, Web, Mobile (Announced)	Unity [14]	<i>In Situ</i> Block-based programming	<i>In Situ</i> Sculpting, Primitives
Zoe [12]	Browse Projects	Subscription	Mobile, VR	Unity	<i>In Situ</i> Templates, Unity Engine Plugin	<i>In Situ</i> SketchFab and Placement/Scaling
Rec Room [9]	Browse	Tokens	Mobile, PC, PC VR, Standalone	Unity	<i>In Situ</i> Node-based Visual Scripting (Circuits V2 [15])	<i>In Situ</i> Sculpting, Primitives
Mozilla Hubs [13]	Client, Browse, Author	Subscription	Web, Mobile VR	AFrame ¹	None	<i>In Situ</i> Web, Import Models
VRChat (Plus) [10]	Browse	Subscription, Merchandise	VR, PC, Mobile	Unity	Node-based Visual Scripting (Unity Plugin)	Unity Plugin
Roblox [11]	Browse	Tokens	Mobile, VR, PC, Consoles	Proprietary	Lua (External)	Roblox Studio
RealityFlow ² [16]	Browse	None	Mobile, VR	Unity	<i>In situ</i> Node-based Visual Scripting	Sculpting, Primitives, Import Models

Drawing from our firsthand experience running an after-school program at a local high school over the past two years, we are cognizant of and preparing for the challenges such as limited internet access and the necessity of safeguarding user privacy in our designs.

We will start by reviewing features from non-VR content creation platforms, then look at existing platforms in terms of their capabilities and approaches to immersive authoring.

2.1 Open-Source Non-VR Platforms

Open-source libraries and platforms that focus on creating content have proliferated through web technologies and communities on the web. Processing [5] is one such library originally developed for Java, but which was later ported to Python and JavaScript (P5.js). It has a web-based editor, p5.js Web Editor³. It includes the capacity to save and share sketches (including any relevant assets) as well as a library of examples.

Twine is an open source hypertext interactive fiction library built on modern web technologies [6]. Twinery.org [6], [7] is a website that embeds the editor and allows users to save projects to local storage without an account, much as Bitsy⁴ does, open to creating and sharing works on its platform. Its open-source nature allows people to add to it for their own development goals. The platform is also designed to be simplistic and easy to understand for novices trying to create works that are experimental or professional coders. This fosters a community of people who create personalized works not often found in higher-funded projects.

These platforms encompass a community of end user creators and developers that contribute to them and are often used in educational contexts due to available tutorials and that they are freely available.

2.2 VR Content Creation Platforms

Many VR multi-user applications are available that can be called immersive platforms or metaverses, as summarized in . We pay particular attention to the affordances of the authoring tools for each platform, their underlying engine, and monetization approach. These combine to shape how users can change content and deploy creations.

Horizon Worlds [8] is the main user-generated content application produced by Meta. Like other platforms, users move between “instances” that can be public or private, many of which

are authored by other users. Content consists of both user-created models as well as scripts built using a block-based programming interface. There is no single “world” where all users interact, such as in MMOGs, and instead interactions are shaped by both instances and parties of users moving between them.

Rec Room [9] “is free, and cross plays on everything from phones to VR headsets. It’s the social app you play like a video game!” As before, it is both a social network and a set of linked instanced scenes. Rec Room supports importing content created using Rec Room Studio, which was developed in partnership with Unity.

VRChat [10] is a cross-platform social application where users can join private or public instances of worlds. Users can upload avatars, worlds, and experiences to the server. There is no immersive authoring except for temporary drawing tools developed by authors within VRChat. The only exporting of content possible is through in game screenshots.

Roblox [11] is a massively popular platform for user-created games, with most of its users on mobile devices and often younger. The base platform mechanics and figures resemble those of other platformer games, and it is one of the few in this paper that has its own proprietary engine. Its potential for educational use cases relies largely on the fact that many students may have played it.

Zoe [12] is a commercially available immersive authoring platform with support for both individual and multi-user changes. It is meant to be a novice creation tool that requires no coding. It has both a standalone program for the Meta Quest 2 and Meta Quest 3 as well as a Unity Plugin with a custom set of components designed to make creating behaviors easier. Content cannot be shared between Unity and the standalone program, however.

Mozilla Hubs, or Hubs by Mozilla [13] is an open-source platform that can be installed on private servers or provisioned through their subscription cloud service. It has features similar to VRChat [10] in supporting multiple users interacting. They have a web authoring tool for creating rooms but focus more on placement of assets created in other tools, and no support for custom scripting.

Each of the tools above require internet connections and were not designed for collecting user data in research studies. Content created in them cannot be exported for further development or publication outside of the platform as well. Finally, all platforms have certain unchangeable aspects, including authoring interfaces, menus and locomotion. Each platform now supports users between 13 and 17 years of age.

¹ <https://aframe.io/>, WebGL and Three.js Based editor

² <https://realityflow.io> , system developed by authors

³ <https://p5js.org/>

⁴ <https://www.bitsy.org/>

3 NEW SOCIAL PLATFORM FEATURES

This section describes the architecture and design of an initial web client that would serve to persist and share user-created content created in the immersive authoring tool. The system was developed by a group of senior design students and consisted of a NodeJS server, a MongoDB database, a React website and a React Native mobile app which both used GraphQL for client-server communication. Users create accounts through the web service and create and share projects that can be accessed through the Unity client. All changes inside the authoring tool are persisted to the database as JSON objects, and projects are associated with rooms managed on a Ubiq [3] server. A simple modeling tool and visual scripting system [16] were developed, though persistence of scripts is still in development at this time.

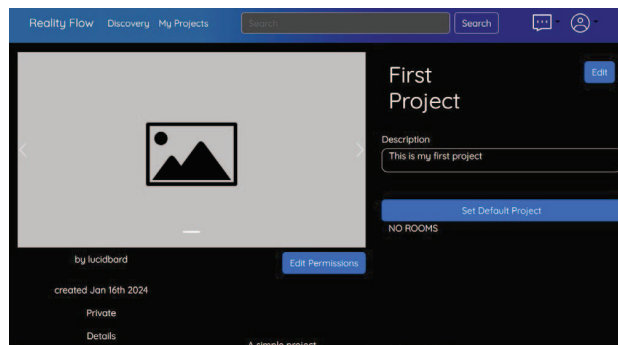


Figure 1: RealityFlow React Web Interface (Project View)

3.1 Authoring Goals and Requirements

Each platform defines content in its own terms, and further constrains how the content is accessed and shared. Rec Room [9] makes a priority, for instance, of building on existing content, while Meta Horizon Worlds [8] has more of a closed approach. When thinking about how open source could benefit future platforms, we consider what is fixed or immutable in these systems. For instance, how can the authoring tools themselves be altered for novices? What types of development is necessary to leave to a desktop editor? How should content be advertised? What limits should be imposed on the modifications end users can make, including changes to UI elements and locomotion? Should core features remain constant, similar to platforms like Rec Room or Horizon Worlds, where numerous essential interactions are common across all experiences? One argument for limiting control customization is the ability to support multiple platforms by default, which both Horizon Worlds and Rec Room do through interfaces customized to desktop, mobile, and VR platforms.

In this section we will explore other social aspects that inform future platform experiments as well as the guiding requirements behind the additions to our platform.

3.2 Roles and Permissions

Giving users the freedom to create content also may create opportunities for them to create harmful or inappropriate content. One approach is to support users flagging content or interactions that could be potentially harmful. They could describe how that project was potentially upsetting or harmful in the report which could be reviewed by a moderator. Bullying is a pernicious and challenging problem that major social media platforms have struggled with for years, and whistleblower, Arturo Bejar, describes one approach that he found successful was to empower users with the tools to report their own perceived harms rather than relying on policies or exclusively external moderation [17]. To

prevent harassment, these reports could lock down these projects to only the author. This can allow users to still author in case of a false report and prevent the spread of harassment if it is genuine until the moderator can discard the project or require a change to its content. For more on moderation in VR, see [18].

In addition to reporting mechanics, due to its collaborative nature co-owners, project access could be managed by both the classroom moderator and the student owner. This is to prevent maliciously or accidental edits. This can also be addressed through more comprehensive save points and access controls.

3.3 Web Client and Features

At a minimum, a web portal supports account management and accessing and browsing content. Content is usually a “scene” (e.g. in Rec Room [9] and Meta Horizon Worlds [8]) which can support multiple instances where users can interact with one another.

Users should be able to launch into experiences directly into their device or set it for starting upon opening a client. Access to projects could be managed both in the client as well as on a web client, editing details and deleting projects.

Publishing projects without a moderation step presents its own risks. Making publishing and sharing projects harder, however, it could make the tool less social and potentially harm the social motivation behind creating experiences.

Making simple experiences that can be extended or accessed is important for both orienting students and showing collaboration features. This could include a range of subjects such as introduction to VR and sickness to more advanced programming of systems. As for user generated content, these projects should be sharable via a code or linked permission. This could allow them to share their work with one another and even make it available through the web. This empowers users to advertise and exchange content instead of being controlled through an algorithm in the world, which many platforms have adopted.

4 CASE STUDY: HIGH SCHOOL PROGRAMS

For the past few years, the authors have been running an after-school enrichment program for a local high school. This has informed the development of the immersive authoring tool. We have found that non-networked VR tools confine users to solitary experiences, which can be a significant limitation in educational processes where students lack collaboration with instructors and peers. Our tool was designed with networking through Ubiq to allow multiple users to work together on the same project in real-time. These are incredibly important to the social dynamics common in secondary and high school age groups.

4.1 Educational Contexts

Educational contexts in general are poorly addressed by current models of social applications for VR. Rec Room is one example where users can be discovered and interact with anyone on the platform and where underaged interactions are assumed to be managed by a parental user or owner of the device.

In educational contexts, if the parent is not the owner of the device, the manager of the device is typically either the school librarian or the program administrator. This means applications must be selected based on the ability to control both the content and potential interactions with other users. As of the authoring of this paper, Meta does not have official support for educational owners or shared usage other than the Meta for Work program [19].

Engaging virtual reality experiences involve social interactions, but some schools have either restricted internet or poor Wi-Fi coverage. Applications that use a local server may provide students with social opportunities without the risk of outsider interactions.

4.2 Student Use Outside of Programs and Classes

A key challenge is the diversity of VR equipment available in schools. Our client targets both VR headsets and mobile devices. Standalone immersive VR authoring applications prevent students from easily picking up work without exporting to files and external storage. We believe storing content on a private server will alleviate this by giving students access to their work on any checked-out device or laptop with access to the server. Also, students may not have access to a high-powered computer, so including a mobile and browser client supports increasing opportunities to participate.

Sufficient tutorials for new users are another key requirement, as students may not be familiar with VR conventions and controls. Although applications such as Oculus First Contact can serve this role, having students orient within the authoring tool to VR could also make creation a part of the initial experience in VR.

One issue when teaching students game development in VR is monitoring students' activities while they are using a headset. We anticipate using Ubiq's [3] new browser support to allow facilitators to view the scenes students are creating and verbally guide them through lessons when needed. This can also researchers to record user behaviors and events alongside traditional telemetry.

Making educational resources that are free and open source is important to ensure the benefits of VR are distributed equally, and that educators can have flexibility in exploring potential use cases both in and out of formal classroom use cases.

5 FUTURE DIRECTIONS

We plan to introduce our system, RealityFlow, to high school students in the future so they can develop collaborative creativity unrestricted by a lack of knowledge, education, or training. Research could explore the long-term impacts of such programs on students' academic and career choices, particularly in STEM fields. We have published the source code for the system on GitHub so other efforts to explore social features and run versions of the workshop we are building. We expect some community servers may be necessary for those unable to run their own servers, but with data wholly owned and controlled by the users.

In the future, we anticipate conducting studies within the three main contexts where a federated content creation platform would most benefit: secondary education contexts, where users would engage with other users' content in an otherwise private server; a college context, where users may wish to collaborate with each other on projects for either competition or further development; and a college classroom where the focus is on pedagogy and design of multiuser interactions.

Another project that has similar goals in terms of keeping data in the hands of users is the Solid project [20] which was initiated by Tim Berners Lee as an effort to address issues around data privacy with current web platforms such as Facebook.

Cross-platform asymmetric authoring will become increasingly important, with users adding content through web browsers and mobile phones at the same time as users interact in VR. Support for users accessing the platforms across multiple platforms is becoming common, but additional work is needed to identify interfaces to support them.

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REFERENCES

- [1] G. Manfredi, U. Erra, and G. Gilio, "A Mixed Reality Approach for Innovative Pair Programming Education with a Conversational AI Virtual Avatar," in *Proceedings of the 27th International Conference on Evaluation and Assessment in Software Engineering*, in EASE '23. New York, NY, USA: Association for Computing Machinery, Jun. 2023, pp. 450–454. doi: 10.1145/3593434.3593952.
- [2] R. Cheng, N. Wu, S. Chen, and B. Han, "Will Metaverse Be NextG Internet? Vision, Hype, and Reality," *IEEE Network*, vol. 36, no. 5, pp. 197–204, Sep. 2022, doi: 10.1109/MNET.117.2200055.
- [3] S. J. Friston *et al.*, "Ubiq: A System to Build Flexible Social Virtual Reality Experiences," in *Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology*, Osaka Japan: ACM, Dec. 2021, pp. 1–11. doi: 10.1145/3489849.3489871.
- [4] "Mastodon." Mastodon GmbH, 2023 2017. Accessed: Jan. 18, 2024. [Online]. Available: <https://github.com/mastodon/mastodon>
- [5] B. Fry and C. Reas, "Processing: programming for the media arts," *AI & Soc.*, vol. 20, pp. 526–538, 2006, doi: 10.1007/s00146-006-0050-9.
- [6] C. Klimas, "Twine." 2009.
- [7] B. Jackson and V. Thompson, "Exploring Twine and Its Use In Active Learning Online," presented at the Society for Information Technology & Teacher Education International Conference, Association for the Advancement of Computing in Education (AACE), Mar. 2018, pp. 1533–1536. Accessed: May 25, 2020. [Online]. Available: <https://www.learntechlib.org/primary/p/182731/>
- [8] "Meta Horizon Worlds." Meta, 2021. [Online]. Available: <https://www.meta.com/horizon-worlds/>
- [9] "Rec Room." Seattle, Washington, USA.
- [10] "VRChat." VRChat Inc., 2023 2014. [Various]. Available: <https://hello.vrchat.com/>
- [11] "Roblox." Roblox Corporation. Accessed: Jan. 15, 2024. [Windows, macOS, iOS, Android, Xbox One, Meta Quest 2, Meta Quest Pro, PlayStation 4]. Available: <https://www.roblox.com/home>
- [12] "Zoe." Zoe Immersive, 2022.
- [13] "Hubs by Mozilla." Mozilla, 2018. [Online]. Available: <https://github.com/mozilla/hubs>
- [14] "Unity." Unity Technologies, 2020. [Online]. Available: <https://unity.com>
- [15] "The Circuits Handbook," Rec Room Developer Blog. Accessed: Apr. 26, 2022. [Online]. Available: <https://blog.recroom.com/posts/2021/5/03/the-circuits-handbook>
- [16] J. T. Murray, "RealityFlow: Open-Source Multi-User Immersive Authoring," in *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, Mar. 2022, pp. 65–68. doi: 10.1109/VRW55335.2022.00024.
- [17] "Written Testimony of Arturo Bejar before the Subcommittee on Privacy, Technology, and the Law, November 7, 2023." Nov. 07, 2023. [Online]. Available: https://www.judiciary.senate.gov/imo/media/doc/2023-11-07_-_testimony_-_bejar.pdf
- [18] N. Sabri, B. Chen, A. Teoh, S. P. Dow, K. Vaccaro, and M. Elshrief, "Challenges of Moderating Social Virtual Reality," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, Hamburg Germany: ACM, Apr. 2023, pp. 1–20. doi: 10.1145/3544548.3581329.
- [19] "VR For Business & Business VR Headsets," Meta for Work. Accessed: Jan. 18, 2024. [Online]. Available: <https://forwork.meta.com/>
- [20] A. V. Samba *et al.*, "Solid: a platform for decentralized social applications based on linked data," *MIT CSAIL & Qatar Computing Research Institute, Tech. Rep.*, 2016, Accessed: Jan. 18, 2024. [Online]. Available: http://emansour.com/research/meccano/solid_protocols.pdf